Written in English for clarity — some technical terms don't translate naturally yet

Link it. Rank it. Trust it.

Smarter Search: Giving AI a Map of Indonesian Law

This dataset moves beyond simple vectors by using a Knowledge Graph to define the relationships between laws, ensuring your AI understands how one regulation connects to another.

Afza Azzindani

Why Simple Search Lacks Legal Trust

- **The Problem**: My initial hybrid search systems (using both keyword and semantic vectors) were fast but incomplete. Retrieval often returns documents that are conceptually correct but legally irrelevant (e.g., an outdated regulation or a low-ranking Perda).

- **The Core Flaw**: The hybrid system prioritized meaning and keywords but lacked trust. A powerful, current constitutional law and an old, superseded regional regulation might be ranked equally if they discuss the same topic.

- **The Solution Imperative**: I realized I needed to structure the legal hierarchy and inject explicit intelligence. This requires Knowledge Graph mapping and Authority/Temporal Scoring to ensure retrieval prioritizes documents based on legal trustworthiness and current status.

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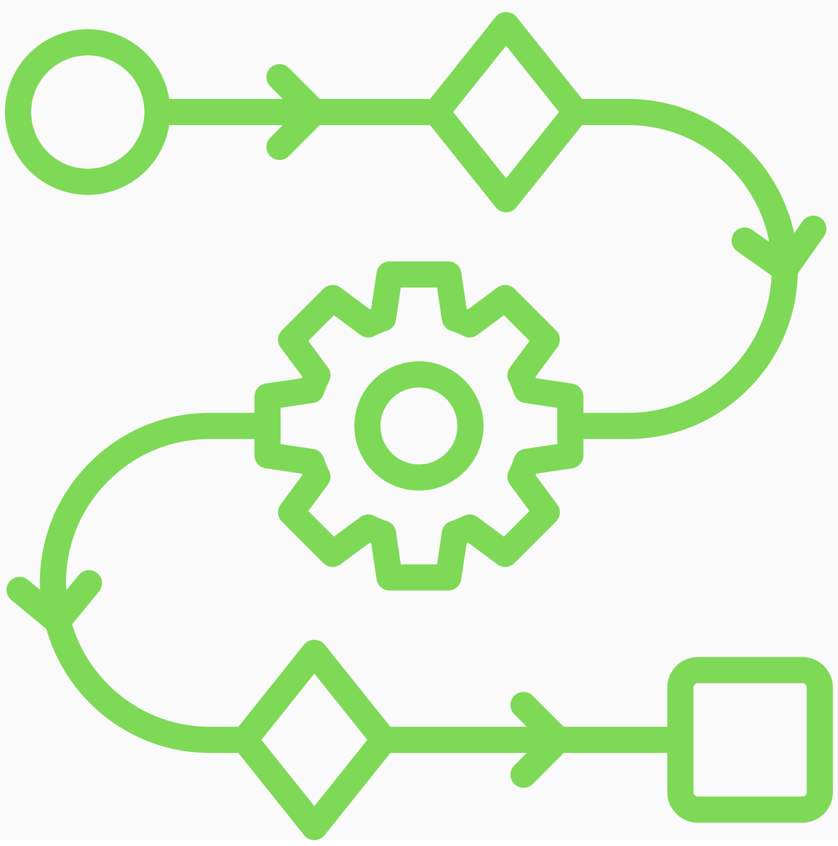
Structuring the Legal Brain

The Knowledge Graph (KG) is the key to adding intelligence and context to legal search, moving beyond basic keyword matching.

- **Analogy:** The Legal Hierarchy Map
    1. Simple Search (Vectors) sees two separate documents: UU No. 12 Tahun 2011 and PP No. 71 Tahun 2019.
    2. The KG sees the relationship: (UU No. 12/2011) ➡️ 'Is Executed By' ➡️ (PP No. 71/2019).
- **How it Works (The Components)**: The KG connects Nodes (things like a specific law or penalty) using Edges (relationships like "defines," "amends," or "cites").
- **Result**: This structure allows the AI to navigate the official legal relationships. When a search finds the UU, the KG ensures the retrieval also includes the legally required implementing document (the PP), providing a complete, citable answer.



Computing Trust and Relevance Scores

The core of the KG enhancement is calculating new metrics that prioritize the legal validity of search results:

Score Field	Purpose (Why it Matters)	Computation (How it's Calculated)	Example
Authority Score	**Ranks legal power.** Filters documents to favor higher regulatory sources	Based on the Indonesian **legal hierarchy**: UUD (highest score) ➡️ UU ➡️ PP ➡️ PERPRES ➡️ PERPROV ➡️ PERDA (lowest score)	UU No. 12/2011 (High Score) is prioritized over a related PERMEN.
Temporal Score	**Ranks freshness.** Ensures the AI cites the most current, active law	Calculated based on the date of enactment (or **last revision**); more recent documents get a higher score	A regulation from 2024 is ranked higher than a related regulation from 1999.
Cross-Ref Strength	**Ranks influence.** Measures how critical a document is to the rest of the legal system	Count of **explicit citations** and mentions by other laws in the corpus	A widely cited UU is ranked higher than a localized, uncited PERDA.

Utility: These scores are used as filters or rerankers in the RAG pipeline, ensuring the **final output is both semantically relevant and legally verifiable**.

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The Intelligence Layer

The Knowledge Graph (KG) integration successfully created a verifiable, research-ready legal resource.

Dataset Summary:

- **Corpus Size:** 10,000+ Indonesian Legal Documents Processed.
- **Processing Success**: ~98.7% success rate, validating the KG construction pipeline.
- **Trust Filter (Authority Score)**: Documents are scored (Avg. 0.72) based on the Indonesian legal hierarchy (UU ➡️ PP ➡️ Perda), enabling AI to prioritize authoritative sources.
- **Relational Density (KG)**: Documents are enriched with an average of 12.4 explicit legal entities and 8.7 concepts for precise graph-based querying.
- **Temporal Score**: The dataset includes the calculated Temporal Score (document freshness) for fine-grained reranking, ensuring up-to-date retrieval.
- **Connectivity (Cross-Ref)**: The cross_ref_strength field quantifies document influence by measuring how often a specific law is cited throughout the corpus.
- **Vector Fidelity**: Preserves both the dense vectors and TF-IDF vectors from the previous pipeline, ready for immediate hybrid search ingestion.
- **Repository**: Azzindani/ID_REG_MD_KG

Utility and Future R&D Roadmap

This Knowledge Graph-enhanced dataset moves your RAG project into advanced, intelligent retrieval:

- **Key Utility**:

  1. **Trustworthy Reranking**: Deploy RAG systems that use the Authority Score and Temporal Score to deliver legally verified answers (the core value).
  2. **Inferential Retrieval**: Use the KG Entities (e.g., sanctions, rights) to perform searches that follow legal relationships and context.
  3. **Legal Analytics**: Cluster documents based on legal concepts and connectivity (Cross-Ref Strength) for deep regulatory analysis.

- **Future Roadmap**:

  1. **LLM Prototype Integration**: Develop a RAG prototype to integrate this vector/KG dataset with my existing fine-tuned LLM, focusing on improving the LLM's citation accuracy during inference.
  2. **Continuous Improvement & Evaluation**: Focus R&D on manually evaluating retrieval failures and designing targeted updates for the Authority and Temporal scoring algorithms.

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This dataset is the public foundation for verifiable Legal AI in Indonesia

How can we best document the scoring algorithms to facilitate easy adoption by other researchers?

Share your experience and help advance open science in legal technology.

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